HMI MONITORING TAG MAP

HMI Tag	PLC Address	Data Type	Access	HMI Display / Function	Reference
Safety_Healthy	M0001	Boolean	Read Only	Safety Ready / Not Ready Status	DWG 011 / 013
Auto_Mode	M0010	Boolean	Read Only	Auto Mode Active Status	DWG 013
Manual_Mode	M0011	Boolean	Read Only	Manual / Setup Mode Active Status	DWG 013
Current_State	M0100~M0114	Boolean State Group	Read Only	S00~S110 Current PLC State Display	DWG 011 / 013
Fault_Hold	M0120	Boolean	Read Only	SF0 Fault / Alarm Hold Status	DWG 011 / 012
Safety_Input_Status	P00, P07	Boolean	Read Only	Safety Healthy / Guard Door Status	DWG 006 / 007
Operator_Input_Status	P01, P02, P03	Boolean	Read Only	Physical Start / Stop / Reset Status	DWG 007
Sensor_Status	P04~P0B	Boolean	Read Only	Infeed / Test / Outfeed / Clamp / Gate Feedback	DWG 007
Process_Output_Status	P20~P25	Boolean	Read Only	Motor / Clamp / PASS / FAIL / RETEST Output Status	DWG 008
Indicator_Output_Status	P26~P29	Boolean	Read Only	Green / Yellow / Red Lamp / Buzzer Status	DWG 008 / 012
Test_Result_Code	D0020	UInt16	Read Only	PASS / FAIL / RETEST / Invalid Result	DWG 011
Test_Complete	D0021	Boolean / UInt16	Read Only	Test Completion Valid Status	DWG 011
Test_Request	D0022	Boolean / UInt16	Read Only	PLC → MCU Test Request Status	DWG 011
MCU_Heartbeat	D0023	UInt16	Read Only	MCU / Modbus Communication Health	DWG 012 / 014
Total_Cycle_Count	D0300:D0301	UInt32 / Counter	Read Only	Total Completed Cycle Count	PLC Sequence
Pass_Count	D0302:D0303	UInt32 / Counter	Read Only	PASS Count	PLC Sequence
Fail_Count	D0304:D0305	UInt32 / Counter	Read Only	FAIL Count	PLC Sequence
Retest_Count	D0306:D0307	UInt32 / Counter	Read Only	RETEST Count	PLC Sequence
Active_Alarm_Code	D0310	UInt16	Read Only	Current Active Alarm Code	DWG 012
Alarm_Count	D0311	UInt16	Read Only	Total Alarm Occurrence Count	DWG 012

NOTE1:
D0300:D0301 TOTAL_CYCLE_COUNT
D0302:D0303 PASS_COUNT
D0304:D0305 FAIL_COUNT
D0306:D0307 RETEST_COUNT
D0310 ACTIVE_ALARM_CODE
D0311 ALARM_OCCURRENCE_COUNT

NOTE2:
PLC–01 SHALL UPDATE
D0300~D0307
ON COMPLETED CYCLE
AND PROVIDE
D0310 / D0311
FOR HMI ALARM DISPLAY.

HMI SETPOINT / COMMAND MAP

HMI Tag	PLC Address	Data Type	Access	Valid Range / Value	PLC Validation	Function
Test_Position_Setpoint	D0050	UInt16	HMI Write / PLC Read	0~4000 mm	Range Check Required	Target Test / Position Setpoint
Motor_Speed_Setpoint	D0051	UInt16	HMI Write / PLC Read	0~100%	0~100 Clamp	Conveyor Motor Speed Setpoint
Process_Alarm_Threshold	D0052	UInt16	HMI Write / PLC Read	Engineering Defined	Range Check Required	Process / Vibration Alarm Threshold
Mode_Select	D0053	UInt16	HMI Write / PLC Read	0 = Auto / 1 = Manual	Safety Ready + No Fault + S10 Ready / Idle	Auto / Manual Mode Request
Proximity_Threshold	D0054	UInt16	HMI Write / PLC Read	0~255	0~255 Clamp	Optional Detection Threshold
HMI_Start_Request	M0200	Boolean	HMI Write / PLC Read	1 = Start Request	Safety Ready + No Fault + S10 Ready / Idle	HMI Start Request
HMI_Stop_Request	M0201	Boolean	HMI Write / PLC Read	1 = Normal Stop Request	Always Accepted	HMI Normal Stop Request
HMI_Alarm_Reset_Request	M0202	Boolean	HMI Write / PLC Read	1 = Alarm Reset Request	Safety Ready + Cause Cleared	HMI Alarm Reset Request
HMI_Alarm_Acknowledge	M0203	Boolean	HMI Write / PLC Read	1 = Alarm Acknowledge	Read Only Alarm History Retained	HMI Alarm Acknowledge
HMI_Count_Reset_Request	M0204	Boolean	HMI Write / PLC Read	1 = Production Count Reset	Manual Mode + HMI Confirmation	Production Counter Reset Request

NOTE1:
HMI COMMAND BITS SHALL BE PROCESSED AS PLC REQUESTS ONLY.
HMI COMMANDS SHALL NOT DIRECTLY WRITE PLC OUTPUTS P20~P29.

NOTE2:
HMI COMMANDS SHALL NOT REPLACE ES–01, DR–01,
SR–01, K1–A, K1–M, OR HARDWIRED SAFETY FUNCTION.

PLC LADDER REQUEST PROCESSING:
SEE DWG 013 / SHEET 3 OF 4 AND SHEET 4 OF 4.

HMI ALARM / HISTORY / DATA LOGGING MAP:
SEE DWG 015 / SHEET 3 OF 3.

NO.	DATE	REVISION
A	2026–09–16	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi–Bin Sorting Cell
SCADA HMI TAG AND COMMAND MAP

DRAWN BY: J. H. KO
DATE: 2026–09–16
DWG NO: 015
SHEET NO: 2 OF 3